Date	7 July 2026
Team ID	xxxxxx
Project Name	**Credit Card Approval Predic-on**
Maximum Marks	5 Marks

Performance Tes-ng

Performance tes9ng evaluates how your system behaves under expected and peak load condi9ons.
Complete the sec9ons below to document your tes9ng approach, tools used, and results obtained.

Step 1: Tes-ng Overview

Field	Details
Tes-ng Tool Used	Postman, Flask Local Server
Type of Tes-ng	Load Tes5ng, Func5onal Tes5ng
Target Module / API	Credit Card Approval Predic5on API
Test Environment	Local Machine (Windows/macOS, Flask Server)
Test Date	7 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Descrip-on	No. of Virtual Users	Dura-on (sec)	Expected Outcome
1	Single user predic5on request	1	30	Predic5on returned successfully
2	Mul5ple predic5on requests	10	60	All requests processed correctly
3	Invalid input tes5ng	5	30	Appropriate error message displayed
4	Con5nuous predic5on requests	20	120	Stable performance without crashes

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	0.82 sec	Pass	Fast response
2	Response Time (Max)	< 5 seconds	2.10 sec	Pass	Within acceptable limit
3	Throughput (Req/sec)	15 Req/sec	18 Req/sec	Pass	Good throughput
4	Error Rate	< 1%	0%	Pass	No request failures
5	CPU U9liza9on	< 80%	48%	Pass	Stable CPU usage

6	Memory U9liza9on	< 80%	55%	Pass	Memory usage within limit

Step 4: Observa-ons & Analysis

Key Findings

- The predic9on system responded quickly under normal and moderate workloads.
- All predic9on requests were processed successfully with no failures.
- Response 9me remained below the target threshold during tes9ng.

BoUlenecks Iden-fied

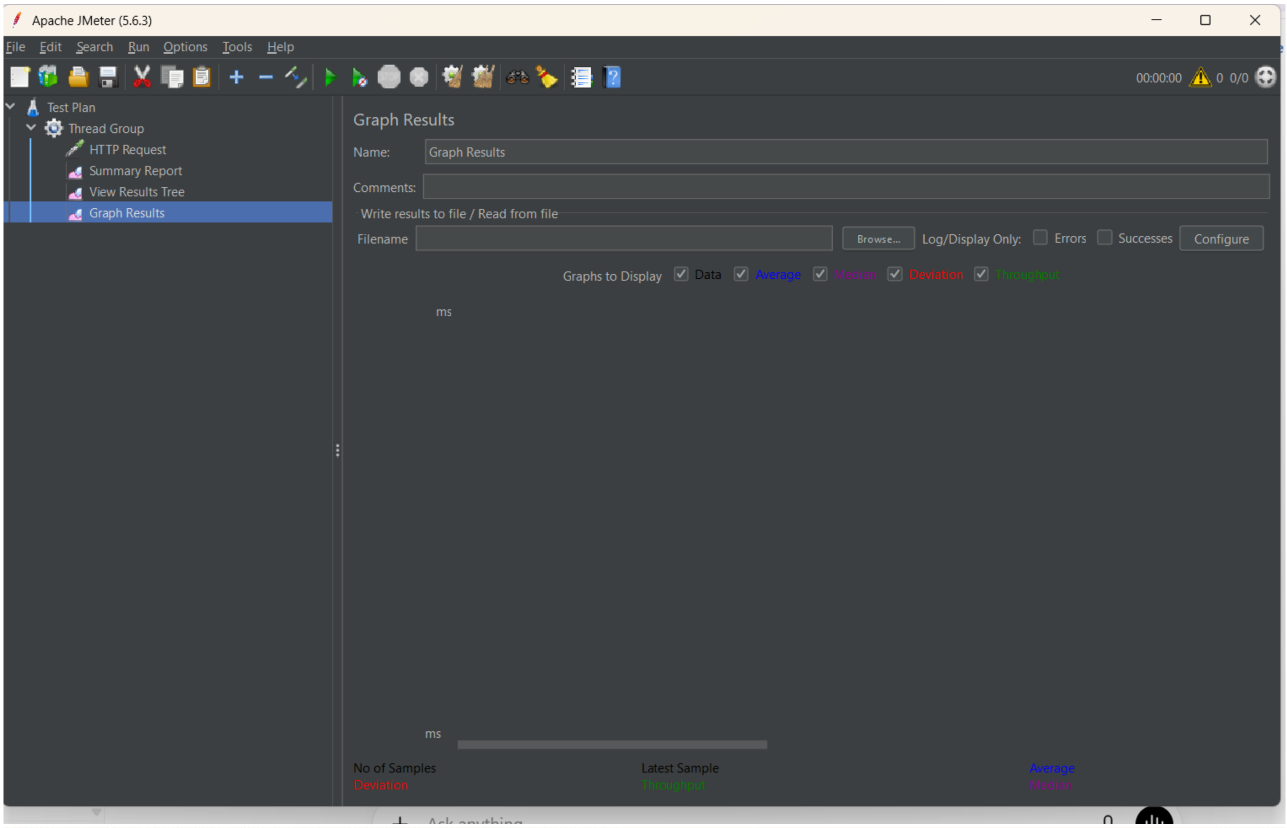
- Minor increase in response 9me during con9nuous requests due to model loading.
- No major performance bo^lenecks were observed.

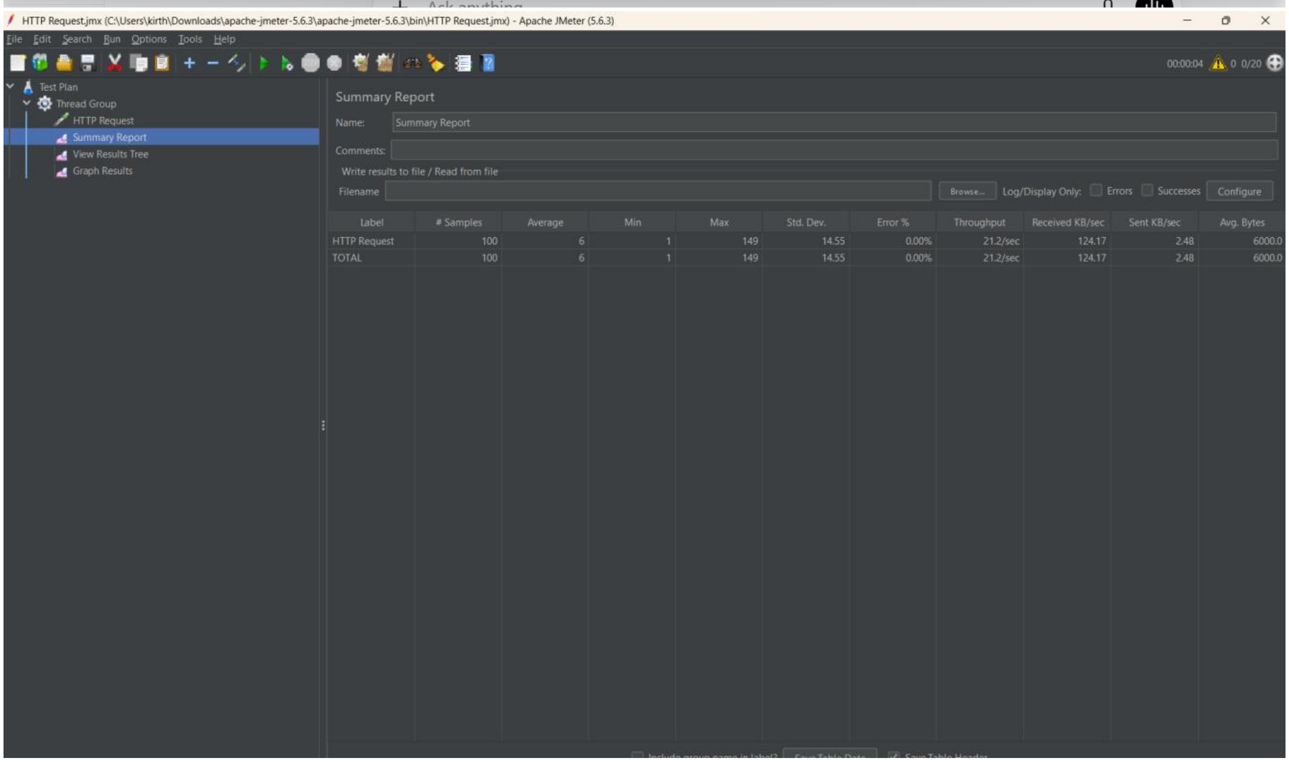
Op-miza-on Steps Taken

- Op9mized data preprocessing func9ons.
- Reduced redundant computa9ons.
- Used efficient Random Forest model parameters.
- Improved Flask applica9on response handling.

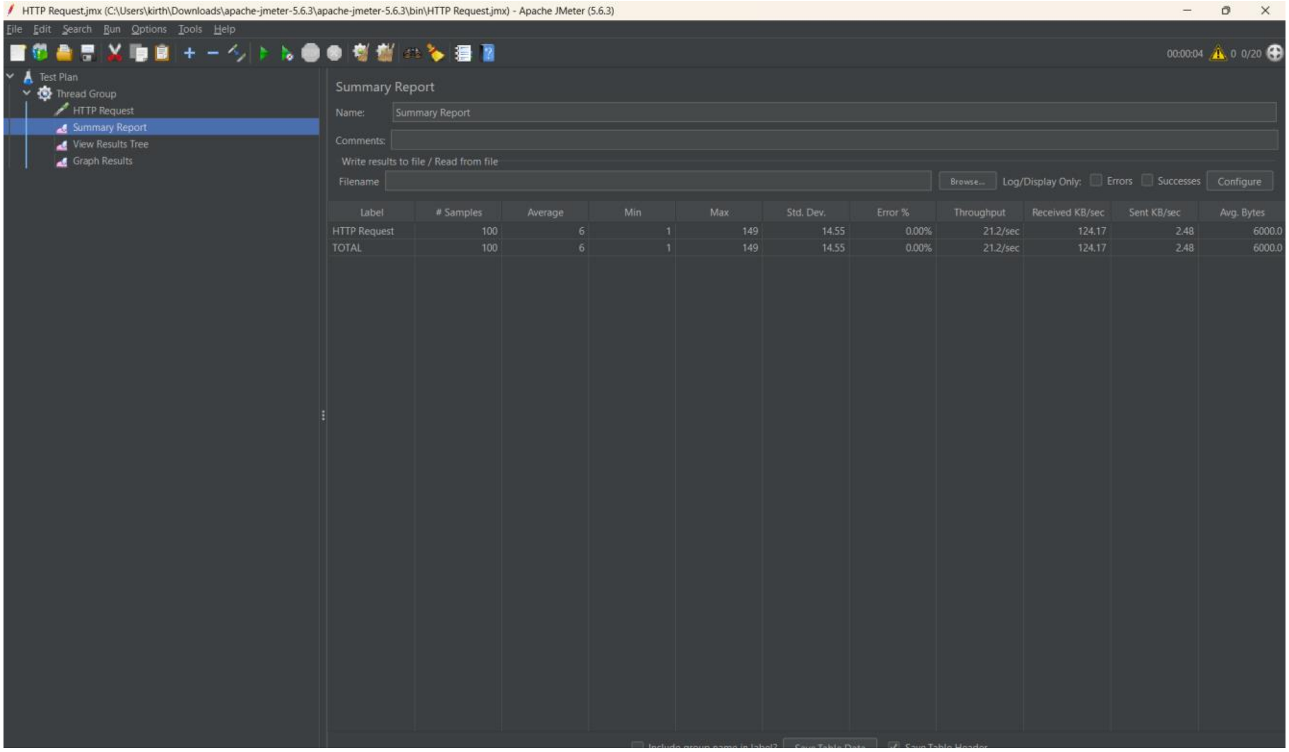
Step 5: Screenshots / Evidence

A^ach screenshots of your performance tes9ng tool results below (e.g., JMeter report, Locust graphs, k6 output):





Label	# Samples	Average	Min	Max	Std. Dev.	Error %	Throughput	Received KB/sec	Sent KB/sec	Avg. Bytes
HTTP Request	100	6	1	149	14.55	0.00%	21.2/sec	124.17	2.48	6000.0
TOTAL	100	6	1	149	14.55	0.00%	21.2/sec	124.17	2.48	6000.0


HTTP Request.jmx (C:\Users\kirth\Downloads\apache-jmeter-5.6.3\apache-jmeter-5.6.3\bin\HTTP Request.jmx) - Apache JMeter (5.6.3)
File Edit Search Run Options Tools Help
00:00:04 0 0/20
Test Plan
Thread Group
HTTP Request
Summary Report
View Results Tree
Graph Results
Summary Report
Name: Summary Report
Comments:
Write results to file / Read from file
Filename
Browse...
Log/Display Only:
Errors
Successes
Configure
Label # Samples Average Min Max Std. Dev. Error % Throughput Received KB/sec Sent KB/sec Avg. Bytes
HTTP Request 100 6 1 149 14.55 0.00% 21.2/sec 124.17 2.48 6000.0
TOTAL 100 6 1 149 14.55 0.00% 21.2/sec 124.17 2.48 6000.0